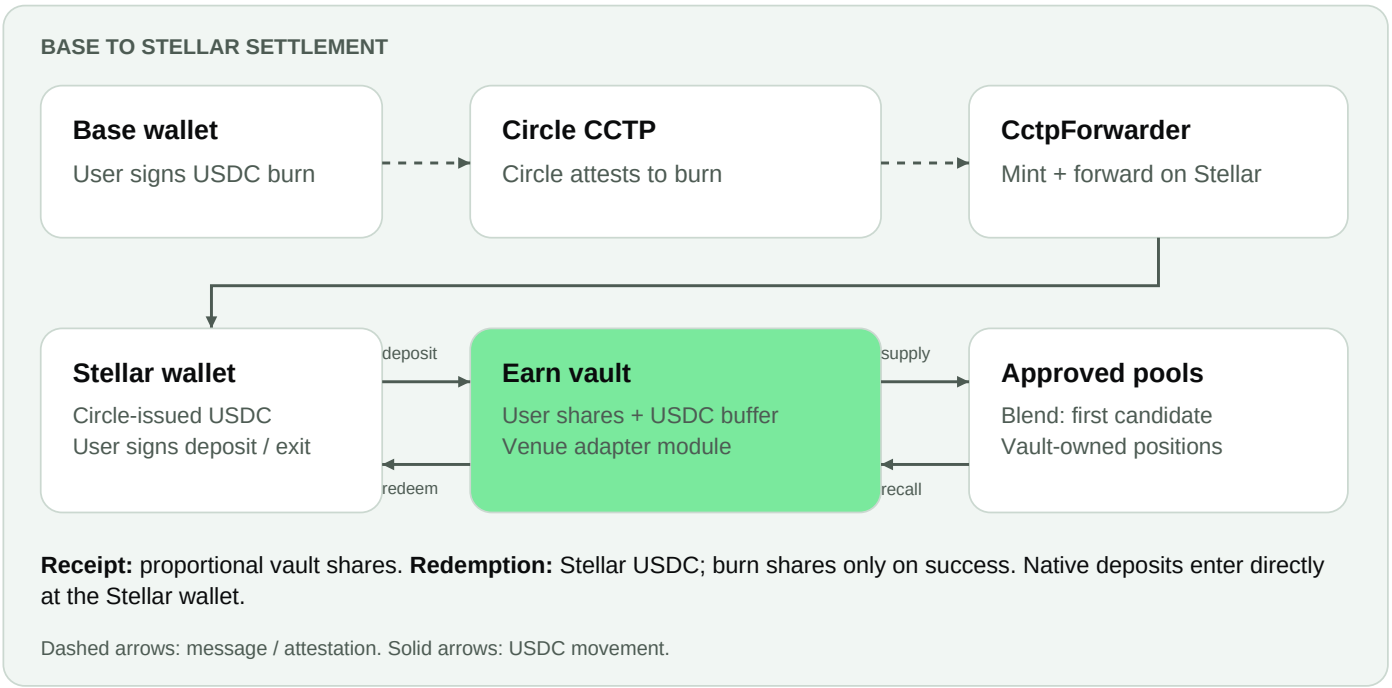
Soroban Automation and Earn

A vault platform powered by Ditto operators: deposit USDC, automate allocation, redeem.

WHY IT MATTERS FOR STELLAR

Reusable vault contracts and operator tooling bring automated yield allocation and recurring onchain activity to Stellar. Earn is Ditto's first Stellar USDC vault; venue adapters support approved lending pools. Base deposits bring external USDC into Stellar.

01 Deposit, allocation and redemption



Receipt: proportional vault shares. **Redemption:** Stellar USDC; burn shares only on success. Native deposits enter directly at the Stellar wallet.

Dashed arrows: message / attestation. Solid arrows: USDC movement.

02 Custody, accounting and authorization

Who holds the assets

Idle USDC is held by the Earn contract; supplied USDC is held in approved lending pools. The vault owns the pool positions; users own vault shares. Neither Ditto nor operators hold funds in a custodial wallet.

Vault entry points

```
deposit(owner, assets, min_shares)
redeem(owner, shares, min_assets)
rebalance(targets)
```

Users authorize deposits / exits. Only the job registry can call rebalance.

Asset and accounting: Circle-issued USDC via its Stellar Asset Contract. Shares track idle USDC plus adapter-valued pool positions, with integer rounding and first-deposit protections. Exits use idle cash, then pool liquidity. No leverage. Future Stellar lending/RWA adapters are outside this award.

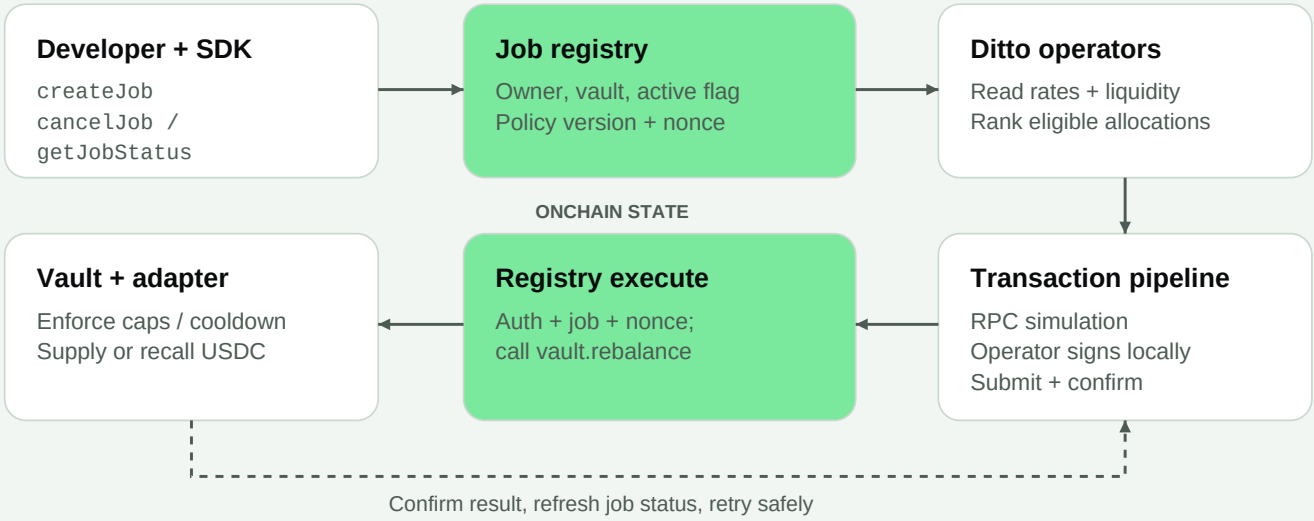
Settlement: Base USDC burn, Circle attestation, CctpForwarder mint-and-forward, then a separate user-authorized vault deposit. Set mintRecipient and destinationCaller to the official forwarder; hook data names the user. Validate the trustline, normalize 6/7 decimals and resume from confirmed steps.

Integration references: Circle CCTP Blend interfaces Stellar contracts

Operators power the platform

Developers deploy compatible vault contracts and use the public SDK to create, cancel and inspect allocation jobs. Earn uses this same infrastructure.

03 Job lifecycle and execution checks



Job validation, nonce update and vault action succeed or revert in one transaction.

04 Allocation and execution controls

Execution	Two executor instances provide continuity. First valid nonce executes; duplicates revert. Confirmed cancellation blocks later calls.
Allocation	Optimize expected net yield after allocation and costs, within exposure caps, available liquidity and buffer limits. Rebalance only above a benefit threshold.
Administration	2-of-3 Stellar account controls executors, caps and pause. Operators cannot redirect withdrawals. Fixed code/pool list; later releases use opt-in migration.
Failure handling	Pause leaves redemptions callable. Illiquid exits revert without burning shares. Monitor failures, balances, transfers and storage TTL; test restoration.

Security: EVM restaking does not automatically secure Stellar execution. Reserve review at kickoff; review and critical/high remediation precede mainnet deposits.

Evidence: publish contract IDs, accounts and transaction hashes. Mainnet acceptance: 5 deposit/exit cycles, 3 test accounts, 2 valid operator actions and 14 monitored days. Separate day-14 pilot targets: 10 external funded wallets and 10,000 USDC external-depositor assets; not guaranteed adoption.

05 Four months to mainnet / $120,000

1 / WEEKS 1-5
Vault template + one pool
Deposit-to-redemption demo

2 / WEEKS 6-10
Operators + SDK + CCTP
Two-pool testnet rebalancing

3 / WEEKS 11-16
Review, fixes, capped MAINNET
14 days of launch monitoring